

DSAA3073: Theories in Data Science

Week 7 Validation Sheet · AdaBoost: Algorithm, Training Certificate, and Limits

Coverage: Week 7

Duration: 30 minutes · **Total:** 130 marks

Answer all questions. For multiple-choice questions, select exactly one answer. Show the essential reasoning requested; unsupported answers may receive no credit.

Part A: Multiple choice

A1 [5 marks]. Which observation, by itself, establishes the population weak-PAC guarantee taught in Week 7?

- (a) A base hypothesis has weighted error 0.40 in one round on one fixed sample.
- (b) Every training round in one run has weighted error below $\frac{1}{2}$.
- (c) For every population distribution in the declared realizable scope and every δ , a polynomial-resource learner achieves error at most $\frac{1}{2} - \gamma$ with probability at least $1 - \delta$, for one fixed $\gamma > 0$.
- (d) The final ensemble has zero error on its training sample.

A2 [5 marks]. At the start of AdaBoost round t , what is D_t ?

- (a) The unknown population distribution on (X, Y) .
- (b) A positive normalized distribution on the indices of the fixed training sample.
- (c) The vector of real ensemble scores.
- (d) The unnormalized exponential factors after round t .

A3 [5 marks]. A binary base hypothesis has current weighted error $\varepsilon_t = \frac{1}{2}$. Which statement matches the registered coefficient formula and its interpretation?

- (a) $\alpha_t = 1$, so the hypothesis receives full weight.
- (b) $\alpha_t = 0$, so the hypothesis adds no signed evidence.
- (c) $\alpha_t < 0$, because the hypothesis is worse than random guessing.
- (d) α_t is the observed test accuracy.

A4 [5 marks]. Which conclusion follows from the Week 7 product theorem without an additional generalization argument?

- (a) $\hat{R}_{G(\text{sign}(F_T))} \leq \prod_t Z_t$.
- (b) Test error decreases at every boosting round.
- (c) Zero training error determines every signed-score magnitude.
- (d) Convexity alone proves a population error bound.

Part B: Definition in use

B1 [20 marks]. In one AdaBoost round, $D_t = (\frac{1}{8}, \frac{1}{8}, \frac{1}{4}, \frac{1}{2})$ and h_t misclassifies exactly indices 1 and 2. Execute the complete round: find ε_t , α_t , Z_t , and D_{t+1} in exact form. Then explain precisely what this successful round establishes about the base learner, what all-round empirical condition would be needed for a fixed-edge training-rate corollary, and why neither statement is the population weak-learning guarantee.

Answer space

Part C: Basic skills

C1 [7 marks]. Let $D_t = (0.20, 0.30, 0.50)$ and suppose only index 1 is misclassified. Compute ε_t , α_t , the three unnormalized next weights, Z_t , and D_{t+1} .

Answer space

C2 [8 marks]. Two valid rounds have $\varepsilon_1 = 0.20$ and $\varepsilon_2 = 0.30$. Using $e^{-0.26} \approx 0.7711$ and $e^{-0.16} \approx 0.8521$, compute: (i) the exact-factor product $\prod_t 2\sqrt{\varepsilon_t(1-\varepsilon_t)}$, (ii) the varying-edge exponential certificate, and (iii) the fixed-edge certificate based on the minimum observed edge. Order the three values from tightest to loosest.

Answer space

Part D: Construction and diagnosis

D1 [17 marks]. A programmer writes the following loop:

1. choose h_t and set ε_t equal to its **unweighted** fraction of mistakes;
2. set $\alpha_t = 1 - \varepsilon_t$;
3. set $D_{t+1}(i) = D_{t(i)} e^{-\alpha_t y_i h_t(x_i)}$;
4. after T rounds, output $F_{T(x)} = \sum_t \alpha_t h_{t(x)}$ as the predicted label.

Repair the loop in the mathematical order required by AdaBoost. For every repair, state the invariant or object type that the defective line violates.

Answer space

D2 [18 marks]. At round 12, two AdaBoost runs on the same training sample both have signed training scores $(0.05, 0.12, 0.40, 0.75, 1.10)$.

Construct two different possible validation-error traces for rounds 12, 13, 14, 15: one in which validation performance improves after round 12 and one in which it deteriorates. Then explain why both traces are compatible with the same zero training error, what the zero-one count discards, and what evidence would be needed for a defensible claim about later test behavior.

Answer space

Part E: Proof and forensic derivation

E1 [25 marks]. An AdaBoost run on four training examples records $D_1 = (\frac{1}{4}, \frac{1}{4}, \frac{1}{4}, \frac{1}{4})$. In round 1 only index 1 is wrong and $\alpha_1 = \frac{1}{2} \log 3$; the log records $Z_1 = \frac{\sqrt{3}}{2}$ and $D_2 = (\frac{1}{2}, \frac{1}{6}, \frac{1}{6}, \frac{1}{6})$. In round 2 only index 2 is wrong and $\alpha_2 = \frac{1}{2} \log 5$. The file records $|Z|_2 = 1$ and labels

$$W_3 = \left(\frac{1}{2\sqrt{5}}, \frac{\sqrt{5}}{6}, \frac{1}{6\sqrt{5}}, \frac{1}{6\sqrt{5}} \right)$$

as the next “distribution.” The signed predictions are

$$(y_i h_1(x_i))_i = (-1, +1, +1, +1), \quad (y_i h_2(x_i))_i = (+1, -1, +1, +1).$$

Write one coherent forensic memo deciding which recorded state is corrupt, recovering the mathematically required state and missing quantity, and issuing justified verdicts on the run’s claimed exact equalities, empirical training certificate, and the legality of starting a third round directly from the stored vector. Your memo must distinguish a numerical coincidence from a theorem whose conditions are actually satisfied.

Answer space

Part F: Synthesis and limitations

F1 [15 marks]. A run produces three pieces of evidence:

1. the stored next-round weights sum to 0.92;
2. differentiating the declared one-dimensional exponential objective gives the logged α_t as its unique minimizer;
3. all training signs are correct, but several signed scores move closer to zero in later rounds.

For each item, choose the useful Week 7 view—algorithmic mechanics, roundwise optimization, or signed-score evolution—and state the strongest conclusion it supports. Then assess the claim: “Together these facts prove that the next round will reduce test error.”

Answer space